

1. The model, pen and paper

1.1. Recursive formulation & optimality conditions

a)	General sequence problem	Our model
Value fct	$v^*(x_0) = \sup_{\{x_{t+1}\}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1})$	$\max_{\{k_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(\underbrace{k_t^\alpha + (1-\delta)k_t - k_{t+1}}_{c_t})$
State var	$x_t \in X$	k_t
Choice var	$x_{t+1} \in \Gamma(x_t)$	k_{t+1}
Feasible set	$\Gamma(x_t)$	$[0, k_t^\alpha + (1-\delta)k_t]$
Reward fct	$F(x_t, x_{t+1})$	$u(c_t) = \begin{cases} \ln(k_t^\alpha + (1-\delta)k_t - k_{t+1}) & \text{if } \gamma=1 \\ \frac{(k_t^\alpha + (1-\delta)k_t - k_{t+1})^{1-\gamma} - 1}{1-\gamma} & \text{else} \end{cases}$
Discount	β	$\beta \in (0, 1)$

The current capital stock k_t is the correct state variable because it determines the value for this period and all future period by defining which consumption-savings bundles are available today given the output produced by the current stock of capital. The SP only hinges on past consumption-savings decisions through capital today - a property that would break if current utility were to depend on last period's consumption level as well, as then we would need c_{t-1} as an additional state variable to be able to compute the value realised today.

$$\begin{aligned}
 \text{b) } v^*(k_0) &= \sup_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(k_t^\alpha + (1-\delta)k_t - k_{t+1}) \\
 &= \sup_{k_1 \in [0, k_0^\alpha + (1-\delta)k_0]} \{ u(k_0^\alpha + (1-\delta)k_0 - k_1) + \beta \sup_{\{k_{t+1}\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^t u(k_t^\alpha + (1-\delta)k_t - k_{t+1}) \} \\
 &= \sup_{k_1 \in [0, k_0^\alpha + (1-\delta)k_0]} \{ u(k_0^\alpha + (1-\delta)k_0 - k_1) + \beta v^*(k_1) \}
 \end{aligned}$$

Rewrite with k' :

$$v(k) = \sup_{k'} \{ u(k^\alpha + (1-\delta)k - k') + \beta v(k') \}$$

$$\text{Constraint 1: } k \geq 0 \rightarrow k_t = 0 \Leftrightarrow \gamma_{t+1} = 0^\alpha = 0 \rightarrow c_{t+1} = 0$$

$$\text{Constraint 2: } c > 0 \rightarrow u_c = \begin{cases} \frac{1}{c} & \xrightarrow{c \rightarrow 0} \infty \\ c^{-\gamma} & \xrightarrow{c \rightarrow 0} \infty \end{cases} \quad \begin{matrix} \text{if } \gamma=1 \\ \text{else} \end{matrix}$$

We see that regardless γ , the marginal utility converges to infinity when consumption converges to 0. Zero consumption can therefore never maximise the value fct. $k_t = 0$ implies zero consumption next period.

c) FOC[k']:
$$V_{k'} = \begin{cases} -\frac{1}{c} + \beta V_{k'}(k') = 0 & \text{if } \gamma=1 \\ \underbrace{-\frac{1}{c^\delta} + \beta V_{k'}(k')}_{u_c} = 0 & \text{else} \end{cases} \Leftrightarrow \frac{1}{\beta c^\delta} = V_{k'}(k')$$

Envelope:
$$v(k) = u(k^\alpha + (1-\delta)k - g(k)) + \beta v(g(k)) \quad \Big| \frac{d}{dk}$$

$$\Leftrightarrow V_k(k) = u_c(c) \cdot (\alpha k^{\alpha-1} + (1-\delta) - g_k(k)) + \beta V_k(g(k)) \cdot g_k(k)$$

$$\Leftrightarrow V_k(k) = \underbrace{(-u_c(c) + \beta V_k(g(k)))}_{=0 \text{ by FOC}} g_k(k) + u_c(c) (\alpha k^{\alpha-1} + (1-\delta))$$

$$\Leftrightarrow V_k(k) = u_c(c) (\alpha k^{\alpha-1} + (1-\delta))$$

Euler eq: plug $V_{k'}(k') = u_c(c) (\alpha k^{\alpha-1} + (1-\delta))$ into FOC[k']:

$$u_c(c) (\underbrace{\alpha k^{\alpha-1} + (1-\delta)}_{\equiv R(k')}) = \frac{1}{\beta c^\delta}$$

$$\Leftrightarrow \beta u_c(c) R(k') = u_c(c)$$

d) (3) gives the intertemporal tradeoff: The marginal utility cost of consuming one less unit today has to equal the marginal utility benefit of saving, i.e. consuming more tomorrow. $R(k')$ expresses the "how much more", i.e. it gives the return on savings in terms of consumption. It is comprised of the increase in output from the additional unit of capital and the capital that will be left next period after depreciation $(1-\delta)$.

1.2. A check that needs no closed form

a) $u_c^{CRRA} = c^{-\delta} \Leftrightarrow c = u_c^{-1/\delta} \equiv u_c^{-1}(\cdot) = c(u_c)$

plugging in:
$$e(k) = 1 - \underbrace{(\beta R(k'))^{-1/\delta} c^{(-\delta \cdot (-1/\delta))}}_{u_c^{-1}(\cdot)} c^{-1}$$

$$= 1 - (\beta R(k'))^{-1/\delta} \frac{c^1}{c}$$

b)
$$e_i = 1 - (\beta \cdot (\alpha k_{gi}^{\alpha-1} + 1 - \delta))^{-1/\delta} \frac{c_{gi}}{c_i}$$